Q07: The Science of Scaling

Power laws, Chinchilla, and the limits

DLS 2026 — Kukla

October 2026

The question

Student question (#21, #22, #39)
Why does scaling work? Is there a "science of scaling," what are its limits, and can learning theory model it?

The empirical finding

$$L(N) \approx L_\infty + (N_0/N)^\alpha$$

- Test loss falls as a **power law** in size N, data D, compute C.
- Holds over many orders of magnitude.
- Lets labs *predict* loss *before* training.

Compute-optimal (Chinchilla)

For fixed $C \approx 6ND$: scale N and D **together**.

- Earlier models: over-parameterized, under-trained.
- Smaller model + more data can beat bigger model + less, at equal cost.

Scaling laws are prescriptive, not just descriptive.

Why would it work?

Partial accounts (no closed theory):

- Prediction as **compression** (lower loss = better world-model).
- **Data-manifold** dimension sets the exponent.
- **Over-parameterization** + SGD find generalizing solutions.

The limits

- **Data wall**: high-quality human text is finite.
- **Diminishing returns**: small exponent $\Rightarrow$ exponential cost.
- **Loss** $\neq$ **capability**; emergence debated.
- **New axes**: post-training + inference-time compute.

Can theory explain it?

- Classical bounds (VC) predict overfitting — too loose.
- Data-manifold / solvable models recover power laws under assumptions.
- No predictive theory matches real exponents across architectures yet.